Six-Month Systems & Formal Verification Synthesis (May - Nov 2026)

8b-is x standardgalactic x peterlodri-sec ? Shared Scientific Report

Target Recognition Format: ACM Software System Award & CAV / POPL Distinguished Artifact

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1. Balanced-Ternary Quantization & Exact Rational Scales (memnet & felix-lang)

- SPQT Base-243 5-Trit Packing: B = sum_{j=0..4} (t_j + 1)*3^j in [0..242] (99.06% bit density)
- Zero-Allocation 4-Register Signal Resolution: 444 B StackMarkdownBuf + 3-Byte SPQT3 [239, 205, 238]
- Exact coprime rational arithmetic (Ratio p/q) with zero floating-point drift across contractions

2. Linear Types, Hardware Capability Monotonicity & SPARK 2022 Verification

- Compile-time linear value checking in felix-lang (single-consumption static guarantee)
- OS-Agnostic PhoenixCapabilityAbiTable across POSIX mmap, WASM linear memory, and ESP32-S3 DMA
- 22 Ada 2022 / SPARK formal packages proved via gnatprove --mode=check_all (0 errors, 0 warnings)
- Self-hosted FIPS 180-4 SHA-256 & constant-time RFC 2104 HMAC-SHA256 webhook verifier

3. Signed Graph Laplacians, Discrete Combinatorics & Statistical Inference

- Harary Structural Balance & Signed Laplacian: x^T L^sigma x = sum (x_u - sigma_uv x_v)^2 = 4*frustrated
- Erdos-Renyi percolation (p_c = ln(N)/N = 0.0004063), Erdos-Kac omega(n), and exhaustive Ramsey R(3,3)=6
- 6-Step Q_6 Statistical Inference Pipeline: coherence = 0.9705, Pearson r = 0.9999, do = 1.1200

4. Shared Scientific Attribution & Rational p/q Signature Map

Contributor	Organization / Spine	Primary Subsystem	p/q
Peter Lodri (peterlodri-sec)	peterlodri-sec / 8b-is	Compiler, Rust/SPARK Kernel, ABI	31/32
Alexandra (ALEX|qMom)	8b-is	Architecture Stewardship & Review	11/9
Chris	8b-is	Acoustic DSP & Hardware Timing	32/27
Dra	8b-is	Operations & E2E Harness	144/89
Nate / Natalie	standardgalactic	memnet Oracle & Library Corpus	243/256

5. Reproducibility & Verification Seal

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Rust Test Suite : 106/106 PASS  |  SPARK Proofs : 22/22 PASS  |  wa-stream E2E : 21/21 PASS
Rational Seal   : P/Q = 44599 / 1424  |  LaTeX SHA-256 : 90cced07ab7c3993d9c2f6cb28009f761a397dff801ae5531d61e
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